higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

(8080205)

10 April 2026 (X-paper)
09:00–12:00

Drawing instruments and non-programmable calculators may be used.

This question paper consists of 8 pages and 1 formula sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL INSTRUMENTS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a new page.
5. Use only a black or blue pen.
6. Write neatly and legibly.

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SECTION A:

QUESTION 1: FLOW MEASUREMENT

1.1 Define the following terms:

1.1.1 Laminar flow

1.1.2 Turbulent flow

1.1.3 Incompressible fluid

1.1.4 Compressible fluid

(4 × 2) (8)

1.2 Study the diagram in FIGURE 1 below, and answer the following questions:

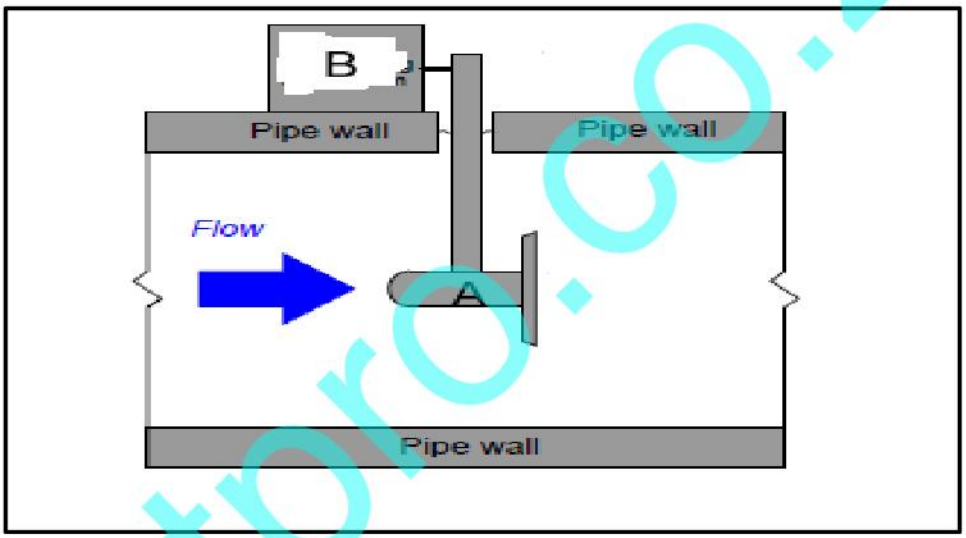


FIGURE 1

1.2.1 Name the flow meter shown above. (1)

1.2.2 Name the parts marked A and B. (2)

1.2.3 Explain the working principle of this flow meter. (4)

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1.3 The following information is known about the TWO weirs in FIGURE 2 below. The width of both weirs is 3 m, and the height is 4 m.

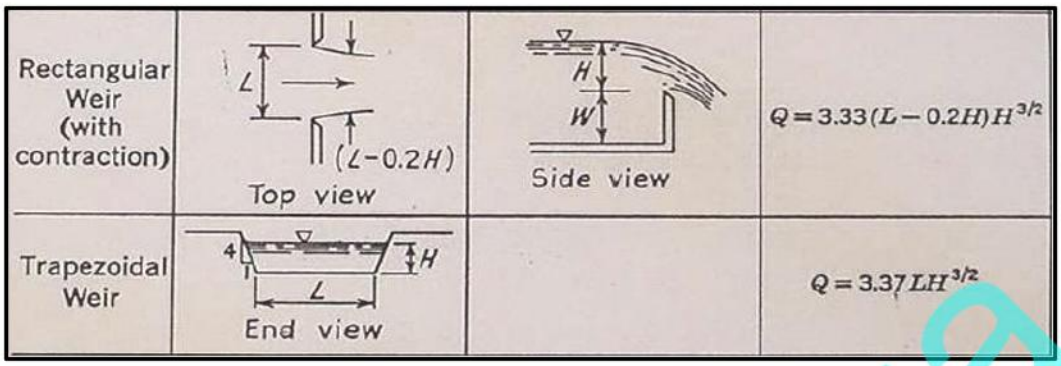


FIGURE 2

1.3.1 Calculate the flow rate of the *rectangular* weir. (2)

1.3.2 Calculate the flow rate of the *trapezoidal* weir. (2)

1.3.3 Comment on the answers. (1)
[20]

QUESTION 2

Bernoulli's equation is an expression of the law of conservation of energy for an inviscid fluid stream. It states the sum total of energy at any point in a passive fluid stream, i.e.:

$$z_1\rho g + \frac{v_1^2\rho}{2} + P_1 = z_2\rho g + \frac{v_2^2\rho}{2} + P_2$$

Where,

z = Height of fluid (from a common reference point, usually ground level)
ρ = Mass density of fluid
γ = Weight density of fluid ($\gamma = \rho g$)
g = Acceleration of gravity
v = Velocity of fluid
P = Pressure of fluid

2.1 When using Bernoulli's equation to develop a precise mathematical relationship between pressure and flow rate in a venturi tube, we will hold FOUR assumptions for the venturi tube system. List these FOUR assumptions. (4)

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2.2 Consider FIGURE 3 below. Take v_1 and v_2 as the flow velocity at points 1 and 2, respectively.

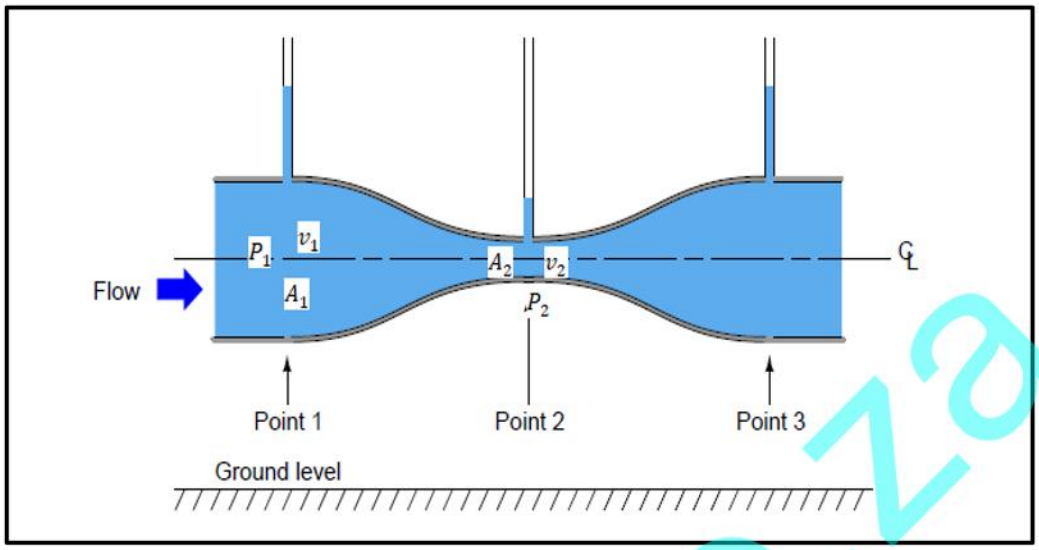


FIGURE 3

Prove that the velocity across point 2 of the venturi tube is given as:

$$v_2 = \sqrt{2}\,\frac{1}{\sqrt{1-\left(\frac{A_2}{A_1}\right)^2}}\sqrt{\frac{P_1 - P_2}{\rho}}$$

(12)
[16]

TOTAL SECTION A: 36

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SECTION B

QUESTION 3: MEASURING DENSITY, HUMIDITY, AND VISCOSITY

3.1 Inductive hydrometers are used to convert the specific weight or density of a liquid into an electrical signal.

3.1.1 Make a neat, labelled sketch of an inductive bridge hydrometer. (6)

3.1.2 Explain the working principle of the meter mentioned in QUESTION 3.1.1 (5)

3.2 FIGURE 4 below is a sketch of a hair hygrometer.

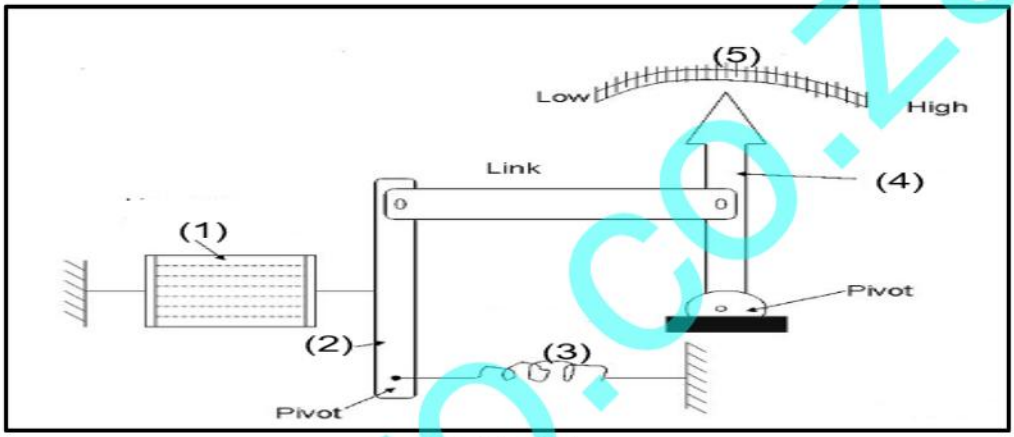


FIGURE 4

3.2.1 Provide labels for the parts marked 1–5. (5)

3.2.2 Explain the working principle of this hygrometer. (4)

3.2.3 What is the requirement for using a hair hygrometer? (1)

3.3 Define the following terms:

3.3.1 Viscosity (2)

3.3.2 Stress (1)

3.3.3 Stokes' law (3)

[27]

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QUESTION 4: MEASURING PH AND CONDUCTIVITY

4.1 Explain the concept of an iso-potential point in a pH measurement using a sketch. (6)

4.2 All materials likely to come into contact with process liquid must be chemically resistant at all measurement temperatures. Name FOUR properties of an ideal housing for these materials. (4)

4.3 Draw a neat, labelled graph of a pH scale covering a range of 0,0 to 14,0. Clearly show acid, neutral, and alkali bases as well as their degrees of strength. (5)

[15]

QUESTION 5: AUTOMATIC CONTROL

5.1 FIGURE 5 below shows a certain control action. Study the sketch, and answer the following questions:

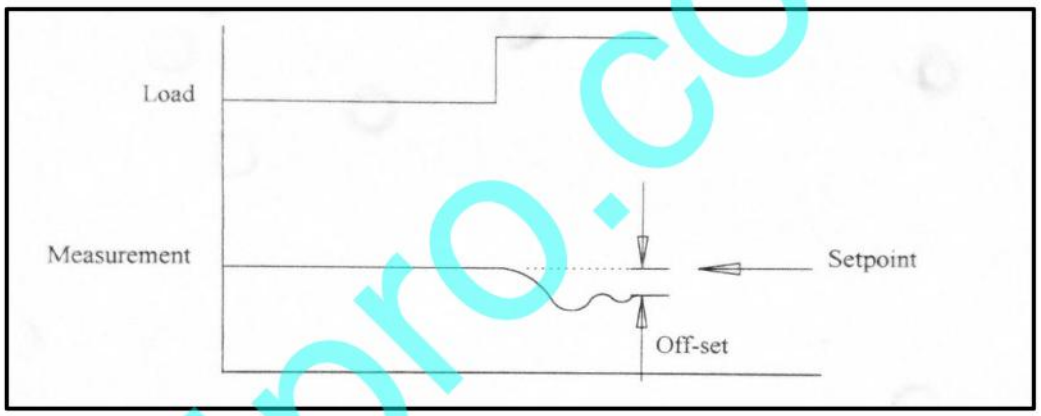


FIGURE 5

5.1.1 Name the type of control action shown in the sketch. (1)

5.1.2 Define the control action shown in the sketch. (2)

5.1.3 What is the disadvantage of this type of control? (2)

5.1.4 Define *offset.* (2)

5.2 Explain the differences between *open-loop control* and *closed-loop control*. (4)

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5.3 FIGURE 6 below shows responses from various control modes. Study the diagram, and answer the following questions:

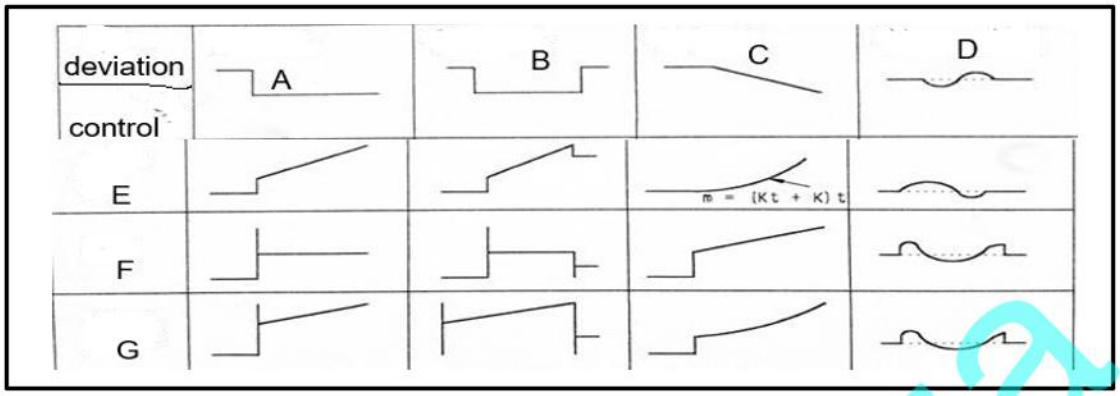


FIGURE 6

5.3.1 Identify the deviations or inputs marked A–D. (4)

5.3.2 Identify the control modes marked E–G. (7)
[22]

TOTAL SECTION B: 64
GRAND TOTAL: 100

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INDUSTRIAL INSTRUMENTS N5

FORMULA SHEET

$$Q = 3{,}33\ (L - 0{,}2\ H)H^{\frac{3}{2}}$$

$$Q = 3{,}37LH^{\frac{3}{2}}$$

$$\boldsymbol{m} = \left(\frac{\boldsymbol{d}}{\boldsymbol{D}}\right)^{\boldsymbol{2}}$$

$$\boldsymbol{E} = \frac{1}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}}$$

$$Q = Av$$

$$R_d = \frac{3{,}54Q\rho}{\mu d}$$

$$Q = 0{,}003998ZC\varepsilon d^2\sqrt{\frac{h}{\rho}}$$

$$Z = Z_R \times Z_D$$

$$\frac{h}{P_{abs}} = \frac{\Delta P}{P_g + P_{atm}} \times 27{,}2$$

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